Zihao Zhang

Master's Student at Fudan University · Advisors: Prof. Yu-Gang Jiang & Prof. Zuxuan Wu
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RESEARCH PROFILE

Researcher at the intersection of **embodied intelligence** and generative visual modeling. I develop controllable and efficient video models for camera-aware scene evolution, large-motion temporal prediction, and human-motion editing—visual foundations for embodied agents. First/co-first author at CVPR 2025 and ACM MM 2026, with seven publications across CVPR, ICCV, ACM MM, SIGGRAPH Asia, and ICASSP.

RESEARCH THEMES

Generative World Models	Spatial & Motion Intelligence	Efficient Visual Generation
Camera- and motion-conditioned video synthesis for modeling how scenes evolve under embodied viewpoints and interactions.	Multimodal spatial reasoning, camera planning, large motion, and temporally coherent scene understanding.	One-step pixel diffusion and lightweight control for scalable, high-resolution visual-dynamics simulation.

EDUCATION

Sep 2023 — Jun 2026 — **Fudan University**
Master's Degree · Shanghai, China
School of Computer Science · GPA: 3.5 · Advisors: Prof. Yu-Gang Jiang & Prof. Zuxuan Wu

Sep 2018 — Jun 2022 — **Wuhan University**
Bachelor's Degree · Wuhan, China
School of Chemistry and Molecular Sciences

AWARDS & HONORS

2025 — **First-Class Academic Scholarship** · Fudan University
2020 & 2021 — **First-Class Scholarship and Grant** · Wuhan University
2025 — **EDEN Patent Application** · Huawei Noah's Ark Lab · Large-motion video frame interpolation

RESEARCH EXPERIENCE

Apr 2025 — Mar 2026 — Research Intern · **BiliBili Inc.**
Virtual Human Group · Image and Video Generation · Shanghai, China

SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation — Co-first author, ACM MM 2026

- Led the one-step pixel-space diffusion framework and designed a progressive multi-stage DiT, Noise-Update-Only Attention, and Drift-aware Timestep Sampling to learn motion, structure, and appearance from coarse to fine.
- Achieved state-of-the-art visual-dynamics prediction: 8.8% lower LPIPS on SNU-FILM, 63.3% faster inference, 10.6% less memory, and up to 51.5% lower LPIPS on 4K benchmarks.

CT-1: Vision-Language-Camera Models for Camera-Controllable Video Generation — Co-first author, ACM MM 2026

- Co-developed a Vision-Language-Camera model that grounds visual observations and language instructions in SE(3) camera trajectories, enabling spatially aware and temporally stable viewpoint planning.
- Contributed to the CT-200K curation pipeline (>47M frames) and controllable video-diffusion system, improving camera-control success by 25.7% and advancing spatially grounded generative world modeling.

Jul 2024 — Mar 2025 — Research Intern · **Huawei**
Noah's Ark Lab · Image and Video Generation · Shanghai, China

EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation — First author, CVPR 2025

- Led EDEN's architecture and training, developing a Transformer Tokenizer, dual-stream temporal attention, and frame-difference conditioning for large, nonlinear motion.
- Reduced LPIPS by nearly 10% on DAVIS and SNU-FILM and by about 8% on DAIN-HD, improving high-fidelity temporal state prediction under fast scene changes.

MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion — Third author, ICCV 2025

- Contributed lightweight pose/reference controllers and two-branch score guidance to transfer target body motion while retaining subject identity, background details, and camera dynamics.
- Reduced GPU memory by approximately 80% versus MotionEditor while improving videos with large camera motion and complex backgrounds, enabling efficient motion-conditioned human-scene synthesis.

Publications

01 **EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation**
Zihao Zhang, Haoran Chen, Haoyu Zhao, Guansong Lu, Yanwei Fu, Hang Xu, Zuxuan Wu
CVPR 2025

02 **SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation**
Zihao Zhang*, Haoyu Zhao*, Siqian Yang, Yidi Wu, Yudong Jiang, Zuxuan Wu
ACM MM 2026

03 **CT-1: Vision-Language-Camera Models Transfer Spatial Reasoning Knowledge to Camera-Controllable Video Generation**
Haoyu Zhao*, **Zihao Zhang***, Jiaxi Gu, Haoran Chen, Qingping Zheng, Pin Tang, Yeying Jin, Yuang Zhang, Junqi Cheng, Zenghui Lu, Peng Shu, Zuxuan Wu, Yu-Gang Jiang
ACM MM 2026

04 **MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion**
Shuyuan Tu, Qi Dai, **Zihao Zhang**, Sicheng Xie, Zhi-Qi Cheng, Chong Luo, Xintong Han, Zuxuan Wu, Yu-Gang Jiang
ICCV 2025

05 **VIDiff: Translating Videos via Multi-Modal Instructions with Diffusion Models**
Zhen Xing, Qi Dai, **Zihao Zhang**, Hui Zhang, Han Hu, Zuxuan Wu, Yu-Gang Jiang
arXiv:2311.18837, 2023

06 **AniME: Adaptive Multi-Agent Planning for Long Animation Generation**
Lisai Zhang, Baohan Xu, Siqian Yang, Mingyu Yin, Jing Liu, Chao Xu, Siqi Wang, Yidi Wu, Yuxin Hong, **Zihao Zhang**, Yanzhang Liang, Yudong Jiang
SIGGRAPH Asia 2025 Posters

07 **EVA: Enhancing Anime Video Generation via Reinforcement Learning**
Yidi Wu, Bingwen Zhu, **Zihao Zhang**, Siqian Yang, Lisai Zhang, Baohan Xu, Mingyu Yin, Yanzhang Liang, Yudong Jiang
ICASSP 2026